

Updates

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I. OPTIMIZATION CHANGE IN DEEP2S DIA IMPLEMENTATION

For the Deep2S attack, I updated how the complex gains A_p are optimized in order to obtain a more stable attack under the same physical amplitude constraint $|A_p| \leq A_{\max}$.

a) Old version (baseline): In the original implementation, I optimized the real and imaginary parts of the gains directly:

$$A_{\text{re}} \leftarrow A_{\text{re}} - \eta_{\text{re}} \nabla_{A_{\text{re}}} \mathcal{L}, \quad A_{\text{im}} \leftarrow A_{\text{im}} - \eta_{\text{im}} \nabla_{A_{\text{im}}} \mathcal{L},$$

using relatively large learning rates. After each gradient step, I enforced the amplitude constraint via a hard projection: for each aperture p , I computed $|A_p|$ and, whenever $|A_p| > A_{\max}$, rescaled A_p back to $|A_p| = A_{\max}$. This procedure caused many entries to sit exactly at the bound $|A_p| = A_{\max}$ and led to oscillatory loss behavior due to the combination of large step sizes and abrupt clipping.

b) New version (current): In the revised implementation, I optimize unconstrained latent variables $Z_{\text{re}}, Z_{\text{im}}$ and map them smoothly to A :

$$A_{\text{re}} = \frac{A_{\max}}{\sqrt{2}} \tanh(Z_{\text{re}}), \quad A_{\text{im}} = \frac{A_{\max}}{\sqrt{2}} \tanh(Z_{\text{im}}),$$

so that the complex gains are

$$A = A_{\text{re}} + j A_{\text{im}}.$$

This guarantees $|A_p| \leq A_{\max}$ by construction (no explicit clipping) and preserves useful gradients near the boundary. Instead of hand-rolled gradient descent, I now use the Adam optimizer on $Z_{\text{re}}, Z_{\text{im}}$, which provides adaptive step sizes and smoother convergence. I retain an ℓ_2 penalty on $|A|$ (through the parameter `lambda_L2`) so that the optimizer does not simply push all gains to their maximum amplitude, but rather uses the available power budget in a more controlled and structured way.

In this revised formulation, the attack loss $\mathcal{L}$ is still computed in the image domain based on A (through the attacked measurements, Deep2S forward model, and MSE to the target volume), but during backpropagation we differentiate with respect to $Z_{\text{re}}, Z_{\text{im}}$. By the chain rule,

$$\frac{\partial \mathcal{L}}{\partial Z_{\text{re}}} = \frac{\partial \mathcal{L}}{\partial A_{\text{re}}} \cdot \frac{\partial A_{\text{re}}}{\partial Z_{\text{re}}} = \frac{\partial \mathcal{L}}{\partial A_{\text{re}}} \frac{A_{\max}}{\sqrt{2}} (1 - \tanh^2(Z_{\text{re}})),$$

$$\frac{\partial \mathcal{L}}{\partial Z_{\text{im}}} = \frac{\partial \mathcal{L}}{\partial A_{\text{im}}} \cdot \frac{\partial A_{\text{im}}}{\partial Z_{\text{im}}} = \frac{\partial \mathcal{L}}{\partial A_{\text{im}}} \frac{A_{\max}}{\sqrt{2}} (1 - \tanh^2(Z_{\text{im}})).$$

The optimizer (Adam in our implementation) then updates $Z_{\text{re}}, Z_{\text{im}}$:

$$Z_{\text{re}} \leftarrow Z_{\text{re}} - \alpha \text{Adam}(\nabla_{Z_{\text{re}}} \mathcal{L}), \quad Z_{\text{im}} \leftarrow Z_{\text{im}} - \alpha \text{Adam}(\nabla_{Z_{\text{im}}} \mathcal{L}),$$

and after each update we recompute $A_{\text{re}}, A_{\text{im}}$ via the $\tanh$ mapping. This guarantees $|A_p| \leq A_{\max}$ by construction, while

still allowing gradients to flow smoothly near the amplitude bound.

Overall, the change is from a “direct + hard-clipping” optimization of A with large steps to a smoothly constrained, Adam-based optimization in the latent space $(Z_{\text{re}}, Z_{\text{im}})$, under the same physical amplitude bound $|A_p| \leq 2$.